

Lecture #26

Date: _____

Stability Analysis

Stability analysis is the study of the numerical method to know whether they are stable or unstable.

A numerical scheme is unstable when, during the solution procedure the result becomes unbounded eventually reaching over flows, independent of choice computing system.

when $|G| \geq 1$ it is unbounded

$|G| \leq 1$ it is bounded

Stability analysis can be performed with Fourier method. A Fourier method is expressed as

$$F_w(i, x) = \omega(x) \exp(i \Omega x_i)$$

where I is imaginary unit,

Ω is Fourier mode,

$\omega(x)$ is mode amplitude.

→ Consistency and stability are necessary and sufficient condition for convergence of an algorithm as the time steps tends to zero.

$$\Delta t \rightarrow 0$$

Convergence

Let U represent the exact solution of a partial differential equation with independent variable x and t , and u represent the approximate

Solution of PDE. Then The finite difference equation is said to be convergent when u tends to at a fixed point U or $|U - u| \rightarrow 0$.

Stability Analysis of Upwind Scheme

Consider the following PDE

$$\frac{\partial f}{\partial t} + c \frac{\partial f}{\partial x} = 0, \quad c > 0 \quad \text{--- (1)}$$

In this scheme, a backward difference in space derivative ($\frac{\partial f}{\partial x}$) is used if $c > 0$ and forward difference is used if $c < 0$.

Since in given equation $c > 0$, So, Replace $\frac{\partial f}{\partial t}$ by 1st order FWD at $P_t(i, j)$ and BWD for ($\frac{\partial f}{\partial x}$), (1) becomes

$$\frac{f_i^{j+1} - f_i^j}{\Delta t} + c \frac{f_i^j - f_{i-1}^j}{\Delta x} = 0$$

$$f_i^{j+1} = f_i^j - \frac{c \Delta t}{\Delta x} (f_i^j - f_{i-1}^j) \quad \text{--- (2)}$$

eq (2) is the upwind scheme for equation (1) for $c > 0$.

In fourier method,

$$f_i^j = G^j e^{ikx}$$

where I is img. mode, k is called wave no. and it is generally real number.

$G = G(k)$ is called amplification factor

the Difference scheme is stable if it is bounded or $|G| \leq 1$ and unstable if it is unbounded or $|G| > 1$

To Find G put

$f_i^j = G^j e^{ikx}$ in eq (2), we have

$$G^{j+1} e^{ikx} = G^j e^{ikx} - \frac{c\Delta t}{\Delta x} (G^j e^{ikx} - G^j e^{ik(x-\Delta x)})$$

let $c\Delta t/\Delta x = \lambda$, then we have

$$G \cdot G^j e^{ikx} = G^j e^{ikx} - \lambda G^j e^{ikx} (1 - e^{-ik\Delta x})$$

$$G \cdot G^j e^{ikx} = G^j e^{ikx} (1 - \lambda(1 - e^{-ik\Delta x}))$$

$$G = 1 - \lambda(1 - (\cos k\Delta x - i \sin k\Delta x))$$

$$= 1 - \lambda(1 - \cos k\Delta x + i \sin k\Delta x)$$

let $k\Delta x = \beta$, so above eq becomes

$$G = 1 - \lambda + \lambda \cos \beta - \lambda i \sin \beta \quad \{i = \sqrt{-1}\}$$

$$|G| = \sqrt{(1 - \lambda + \lambda \cos \beta)^2 + (-\lambda \sin \beta)^2}$$

$$|G| = \sqrt{(1 - \lambda)^2 + \lambda^2 + 2(1 - \lambda)(\lambda \cos \beta) + \lambda^2}$$

$$|G| = \sqrt{(1 - \lambda)^2 + 2\lambda(1 - \lambda)(\cos \beta) + \lambda^2 \cos^2 \beta + \lambda^2 \sin^2 \beta}$$

$$= \sqrt{(1 - \lambda)^2 + 2(1 - \lambda)\lambda \cos \beta + \lambda^2 (\cos^2 \beta + \sin^2 \beta)}$$

$$= \sqrt{(1 - \lambda)^2 + \lambda^2 + 2(1 - \lambda)\lambda \cos \beta}$$

$$= \sqrt{1 - 2\lambda + \lambda^2 + \lambda^2 + 2(1 - \lambda)\lambda \cos \beta}$$

$$= \sqrt{1 + 2\lambda^2 - 2\lambda + 2(1 - \lambda)\lambda \cos \beta}$$

Since $|G| \leq 1$, so

$$\sqrt{1 + 2\lambda^2 - 2\lambda + 2(1 - \lambda)\lambda \cos \beta} \leq 1$$

Squaring both sides of above eq, we have

$$1 + 2\lambda^2 - 2\lambda + 2\lambda \cos\beta - 2\lambda^2 \cos\beta \leq 1$$

$$2\lambda(\lambda - 1 + \cos\beta - \lambda \cos\beta) \leq 1 - 1$$

$$\lambda^2(1 - \cos\beta) - \lambda(1 - \cos\beta) \leq 0$$

$$\lambda^2(1 - \cos\beta) \leq \lambda(1 - \cos\beta)$$

$$\lambda \leq 1$$

$\Rightarrow |Q| \leq 1$ So, it is stable.

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Example #20

Find the stability condition of the following numerical scheme by applying Fourier method.

$$u_i^{j+1} = C^2 \lambda^2 u_{i-1}^j + 2(1 - C^2 \lambda^2) u_i^j + C^2 \lambda^2 u_{i+1}^j - u_i^{j-1}$$

Sol

Put $G^j e^{Ikx} = u_i^j$ in given eq we have

$$G^{j+1} e^{Ikx} = C^2 \lambda^2 G^j e^{Ik(x-\Delta x)} + 2(1 - C^2 \lambda^2) G^j e^{Ikx} + C^2 \lambda^2 G^j e^{Ik(x+\Delta x)} - G^{j-1} e^{Ikx}$$

$$G \cdot G^j e^{Ikx} = G^j e^{Ikx} (C^2 \lambda^2 e^{-Ik\Delta x} + 2(1 - C^2 \lambda^2) + C^2 \lambda^2 e^{Ik\Delta x} - G^{-1})$$

$$G = C^2 \lambda^2 (e^{Ik\Delta x} + e^{-Ik\Delta x}) + 2(1 - C^2 \lambda^2) - G^{-1}$$

$$G = 2C^2 \lambda^2 \left(\frac{e^{Ik\Delta x} + e^{-Ik\Delta x}}{2} \right) + 2(1 - C^2 \lambda^2) - G^{-1}$$

Let $Ik\Delta x = \beta$ and $\left\{ \therefore \frac{e^\beta + e^{-\beta}}{2} = 1 - \cos\beta \right\}$

So,

$$G = 2C^2 \lambda^2 (1 - \cos\beta) + 2(1 - C^2 \lambda^2) - G^{-1}$$

$$G = 2c^2\lambda^2 \cos\beta + 2 - 2c^2\lambda^2 - G^{-1}$$

$$G - 2c^2\lambda^2 \cos\beta - 2 + 2c^2\lambda^2 + G^{-1} = 0$$

$$G + 2c^2\lambda^2 (1 - \cos\beta) + G^{-1} - 2 = 0$$

$$(\therefore 1 - \cos\beta = 2 \sin^2 \beta/2)$$

So,

$$G + 2c^2\lambda^2 (2 \sin^2 \beta/2) + G^{-1} - 2 = 0$$

$$G + 2(2c^2\lambda^2 \sin^2 \beta/2 - 1) + G^{-1} = 0$$

Multiply above eq by G .

$$G^2 + 2(2c^2\lambda^2 \sin^2 \beta/2 - 1)G + 1 = 0$$

$$a = 1, \quad b = 2(2c^2\lambda^2 \sin^2 \beta/2 - 1), \quad c = 1$$

which is Quadratic equation.

$$G = \frac{-b \pm \sqrt{b^2 - 4ac}}{2}$$

$$G = \frac{-4(2c^2\lambda^2 \sin^2 \beta/2 - 1) \pm \sqrt{[2(2c^2\lambda^2 \sin^2 \beta/2 - 1)]^2 - 4(1)(1)}}{2}$$

$$= \frac{2 - (2c^2\lambda^2 \sin^2 \beta/2 + 1) \pm \sqrt{[2c^2\lambda^2 \sin^2 \beta/2 + 1]^2 - 1}}{2}$$

$$G = -(2c^2\lambda^2 \sin^2 \beta/2 + 1) \pm \sqrt{4c^4\lambda^4 \sin^4 \beta/2 + 4c^2\lambda^2 \sin^2 \beta/2 - 1}$$

$$G = -2c^2\lambda^2 \sin^2 \beta/2 + 1 \pm 2c\lambda \sin \beta/2 \sqrt{c^2\lambda^2 \sin^2 \beta/2 - 1}$$

There are three cases

$$\textcircled{1} \quad c^2\lambda^2 \sin^2 \beta/2 > 1$$

$$\textcircled{2} \quad c^2\lambda^2 \sin^2 \beta/2 < 1$$

$$\textcircled{3} \quad c^2\lambda^2 \sin^2 \beta/2 = 1$$

Case 1

$$G = 1 - 2c^2\lambda^2 \sin^2 \beta/2 \pm i 2c\lambda \sin \beta/2 \sqrt{1 - c^2 \sin^2 \beta/2} (\lambda^2)$$

Taking square both sides after taking mod

$$(|G|)^2 = \left(\sqrt{(1 - 2c^2\lambda^2 \sin^2 \beta/2)^2 + (2c\lambda \sin \beta/2 \sqrt{1 - c^2 \sin^2 \beta/2})^2} \right)^2$$

$$\begin{aligned} |G|^2 &= (1 - 2c^2\lambda^2 \sin^2 \beta/2)^2 + 4c^2\lambda^2 \sin^2 \beta/2 (1 - c^2 \sin^2 \beta/2) \\ &= 1 + 4c^4\lambda^4 \sin^4 \beta/2 - 4c^2\lambda^2 \sin^2 \beta/2 + 4c^2\lambda^2 \sin^2 \beta/2 \\ &\quad - 4c^4\lambda^4 \sin^4 \beta/2 \end{aligned}$$

$$|G|^2 = 1$$

Scheme is stable in this case.

Case 2

$$G = 1 - 2c^2\lambda^2 \sin^2 \beta/2 + 2c\lambda \sin \beta/2 \sqrt{c^2 \lambda^2 \sin^2 \beta/2 - 1}$$

$$|G| = |1 + 2c\lambda \sin \beta/2 (\sqrt{c^2 \lambda^2 \sin^2 \beta/2 - 1} - c\lambda \sin \beta/2)|$$

$$|G| > 1$$

So, scheme is unstable.

Case 3

$$\text{If } c^2 \lambda^2 \sin^2 \beta/2 = 1$$

$$\Rightarrow |G| = |-1|$$

$$|G| = 1$$

So, scheme is stable.

Example # 21

Check the stability of FWD method for Heat equation by Fourier method.

Sol

Since 1D heat eq is

$$\frac{\partial u}{\partial t} = \alpha^2 \frac{\partial^2 u}{\partial x^2} \quad u(x, t)$$

Apply FWD on R.H.S and FD on L.H.S.

at above eq. we have,

$$\frac{u_{i,j+1} - u_{i,j}}{\Delta t} = \alpha^2 \left(\frac{u_{i-1,j} - 2u_{i,j} + u_{i+1,j}}{h^2} \right)$$

or k

$$u_{i,j+1} = u_{i,j} + \frac{\alpha^2 k}{h^2} (u_{i-1,j} - 2u_{i,j} + u_{i+1,j})$$

$$u_{i,j+1} = u_{i,j} + \lambda (u_{i-1,j} - 2u_{i,j} + u_{i+1,j})$$

$$\lambda = \alpha^2 k / h^2 \quad \text{put } G^j e^{ikx} = u_i^j$$

we, have

$$u_{i,j+1} = \lambda u_{i-1,j} + (1-2\lambda) u_{i,j} + \lambda u_{i+1,j}$$

$$G^{j+1} e^{ikx} = \lambda G^j e^{ik(x-\Delta x)} + (1-2\lambda) G^j e^{ikx} + \lambda G^j e^{ik(x+\Delta x)}$$

$$G \cdot G^j e^{ikx} = G^j e^{ikx} (\lambda e^{-ik\Delta x} + (1-2\lambda) + \lambda e^{ik\Delta x})$$

$$G = \lambda e^{-\beta} + (1-2\lambda) + \lambda e^{\beta} \quad \therefore \beta = ik\Delta x$$

$$G = \lambda (e^{-\beta} + e^{\beta}) + (1-2\lambda)$$

$$G = 2\lambda \cos \beta + 1 - 2\lambda \quad \therefore e^{-\beta} + e^{\beta} = 2 \cos \beta$$

$$G = 1 - 2\lambda (1 - \cos \beta) \quad \therefore 1 - \cos \beta = 2 \sin^2 \beta/2$$

$$G = 1 - 2\lambda (2 \sin^2 \beta/2)$$

$$G = 1 - 4\lambda \sin^2 \beta/2$$

$$|G| = |1 - 4\lambda \sin^2 \beta/2| < 1$$

$$|G| \leq 1$$

Since This FWD scheme is unconditional stable.

Lecture #28

Example #22

Check the stability of BWD method for heat equation using Fourier method.

Solution

Since 1D heat eq

$$\frac{\partial u}{\partial t} = \alpha^2 \left(\frac{\partial^2 u}{\partial x^2} \right) \quad u(x, t)$$

Now apply BWD on R.H.S and FD on L.H.S. of heat eq

$$\frac{u_{i,j} - u_{i,j-1}}{k} = \alpha^2 \left(\frac{u_{i-1,j} - 2u_{i,j} + u_{i+1,j}}{h^2} \right)$$

$$u_{i,j-1} = u_{i,j} - \alpha^2 k / h^2 (u_{i-1,j} - 2u_{i,j} + u_{i+1,j})$$

$$u_{i,j-1} = u_{i,j} - \lambda (u_{i-1,j} - 2u_{i,j} + u_{i+1,j})$$

$$\therefore \lambda = \alpha^2 k / h^2$$

$$u_{i,j-1} = u_{i,j} - \lambda u_{i-1,j} + 2\lambda u_{i,j} - \lambda u_{i+1,j}$$

$$u_{i,j-1} = -\lambda u_{i-1,j} + (1+2\lambda) u_{i,j} - \lambda u_{i+1,j}$$

Now put $u_{i,j} = G^j e^{ikx}$

$$G^{j-1} e^{ikx} = -\lambda G^j e^{ik(x-\Delta x)} + (1+2\lambda) G^j e^{ikx} - \lambda G^j e^{ik(x+\Delta x)}$$

$$G^{-1} G^j e^{ikx} = G^j e^{ikx} (-\lambda e^{-ik\Delta x} + (1+2\lambda) - \lambda e^{ik\Delta x})$$

$$G^{-1} = -\lambda e^{-\beta} - \lambda e^{\beta} + (1+2\lambda)$$